

1. Swap the **left** and **right** halves of an image.
2. Swap the **top** and **bottom** halves of an image.
3. Create a **horizontal mirror image** using NumPy slicing.
4. Create a **vertical mirror image** using NumPy slicing.
5. Divide an image into **9 equal blocks** and display each block separately.
6. Display only the **Blue, Green, and Red** channels as separate grayscale images.
7. Crop the **center 200 × 200** region, resize it to **600 × 600**, and display both images.
8. Resize an image to **25%, 50%, and 200%** of its original size and compare the outputs.
9. Change an entire column to Green.
10. Change an entire row to Blue.
11. Draw a **red border** around the image using pixel manipulation.
12. Crop the **bottom-right 250 × 250** region.
13. Crop the **top-left 200 × 200** region of an image.
14. Draw a **green square** using pixel modification.
15. Crop all **four quadrants** of an image.

USE CASES

OpenCV Laboratory – Problem Statements with Use Cases

1. Passport Photo Preparation

Problem Statement:

A passport application portal accepts photographs of only **600 × 600 pixels**. Write a program to read a passport photo, resize it to the required dimensions, and save the resized image.

Concepts Used:

- Image loading
- Image resizing
- Image saving

2. Social Media Profile Picture

Problem Statement:

A social media platform displays only the **center portion** of a profile picture. Write a program to crop the **center 300 × 300 pixels** of an image and save it as the profile picture.

Concepts Used:

- Cropping
 - Image saving
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3. CCTV Face Extraction

Problem Statement:

A CCTV image contains a person standing at the center of the frame. Extract only the **Region of Interest (ROI)** containing the person's face by cropping the center of the image.

Concepts Used:

- Image cropping
- ROI

4. Medical Image Preprocessing

Problem Statement:

Medical imaging systems often process grayscale images to reduce computational complexity. Write a program to convert a color medical image into grayscale and save the output.

Concepts Used:

- Color conversion
- Grayscale
- Image saving

5. Traffic Signal Detection

Problem Statement:

Autonomous vehicles often detect traffic lights using the HSV color space. Convert a traffic image from BGR to HSV format.

Concepts Used:

- HSV conversion

6. Printing Image Information

Problem Statement:

Before processing an image, an engineer wants to know its dimensions, color channels, and data type. Write a program to display all these properties.

Concepts Used:

- Image properties
- Shape

- Data type

7. Pixel Inspection

Problem Statement:

A scientist wants to examine the color value of a specific location in a satellite image. Write a program to display the BGR values of pixel (150,150).

Concepts Used:

- Pixel access

8. Highlighting a Defective Area

Problem Statement:

In a quality inspection system, a defective region needs to be highlighted. Draw a **100 × 100 green square** on the image by modifying pixel values.

Concepts Used:

- Pixel modification

9. Image Compression Preview

Problem Statement:

A website generates thumbnails before uploading images. Resize the original image to **25%**, **50%**, and **75%** of its original size and compare the outputs.

Concepts Used:

- Image resizing

10. Compare Image Quality

Problem Statement:

Different resizing techniques produce different image qualities. Resize an image using **Nearest Neighbor**, **Linear**, and **Cubic interpolation**, then compare the results.

Concepts Used:

- Interpolation

11. Image Dataset Preparation

Problem Statement:

A machine learning model requires images of size **224 × 224 pixels**. Resize an image accordingly and save it for model training.

Concepts Used:

- Image resizing
- Image saving

12. Wildlife Monitoring

Problem Statement:

A wildlife camera captures large images, but the animal appears only in the lower-right corner. Crop only the bottom-right region and display it.

Concepts Used:

- Cropping

13. Document Scanning

Problem Statement:

A scanned document contains unnecessary borders. Crop only the useful portion of the document before saving it.

Concepts Used:

- Cropping
- Image saving

14. Digital Photo Album

Problem Statement:

Convert a color photograph into grayscale to create a black-and-white photo album.

Concepts Used:

- Grayscale conversion

15. RGB Image Display

Problem Statement:

Matplotlib displays OpenCV images with incorrect colors because OpenCV stores images in BGR format. Convert the image to RGB before displaying it.

Concepts Used:

- RGB conversion

16. Fruit Sorting System

Problem Statement:

A fruit sorting machine identifies ripe fruits using color information. Convert the captured image into HSV color space before further processing.

Concepts Used:

- HSV conversion

17. Human Skin Detection

Problem Statement:

Many face detection systems use the YCrCb color space for skin detection. Convert an image into YCrCb format.

Concepts Used:

- YCrCb conversion

18. Photo Enhancement

Problem Statement:

An image enhancement application processes brightness separately from color information. Convert the image into LAB color space.

Concepts Used:

- LAB conversion

19. Security Camera Analysis

Problem Statement:

A surveillance image is divided into four equal sections so that each region can be analyzed independently. Divide the image into four quadrants and display them.

Concepts Used:

- Cropping
- ROI

20. Panorama Processing

Problem Statement:

An image stitching application processes the left and right halves of an image separately. Extract both halves and display them.

Concepts Used:

- Cropping

21. Agricultural Drone Imaging

Problem Statement:

A drone captures high-resolution crop images. Extract only the upper half of the image for crop health analysis.

Concepts Used:

- Cropping

22. Satellite Image Analysis

Problem Statement:

Divide a satellite image into **9 equal blocks** so that each block can be processed independently.

Concepts Used:

- Cropping
- Image partitioning

23. Image Reflection

Problem Statement:

A photo editing application provides a mirror effect. Create both horizontal and vertical mirror images of the given photograph.

Concepts Used:

- NumPy slicing
- Image manipulation

24. AI Data Augmentation

Problem Statement:

To increase the diversity of training data for a deep learning model, generate horizontally and vertically flipped versions of an image.

Concepts Used:

- Image flipping
- Data augmentation